BEYOND BINARY: TESTING A GRADUATED PIOTROSKI F_SCORE IN INDIA'S NSE 100 UNIVERSE

Submitted in partial fulfilment of the requirements for the award of the degree of

MASTER OF BUSINESS ADMINISTRATION (MBA)

By

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March 2026

DECLARATION

I hereby declare that the dissertation titled "Beyond Binary: Testing a Graduated Piotroski F_Score in India's NSE 100 Universe" submitted to **Sri Sathya Sai Institute of Higher Learning** in partial fulfilment of the requirements for the award of the degree of **Masters of Business Administration (MBA)** is a record of original work carried out by me under the guidance of **Sri Rajeev Rajan, Assistant Professor, Department of Management & Commerce.**

I further declare that this work has not been submitted for the award of any degree or diploma in any university or institution.

Place: Bengaluru, Karnataka

Date: March 2026

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CERTIFICATE

This is to certify that the dissertation titled "Beyond Binary: Testing a Graduated Piotroski F_Score in India's NSE 100 Universe" is a bona fide research work carried out by Shree. **S Adithya, Registration Number 24030302038,** in partial fulfilment of the requirements for the award of the **MBA Degree**, under my guidance and supervision.

Place: Bangalore

Date: 21st March 2026

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ABSTRACT

Value investors have long used the book-to-market ratio to identify potentially underpriced stocks, but not all cheap stocks are worth investing in. Some are genuinely priced lower because of their deteriorating financial health. A nine point signal checklist was developed in a study by (Piotroski, 2000), based on publicly available financial statements, to separate value stocks that are improving from deteriorating ones. The study asks two questions: (a) Does the F_Score work in India's NSE 100, and (b) Can it be made more sophisticated to make it even better?

A total of 189 unique companies were covered in the study, over a period of 12 years from July 2013 to June 2025. Each year, the top 30% of stocks by book-to-market ratio are identified as value stocks. Two models are then tested on this identical sample. Model 1 applies Piotroski's original binary scoring framework, while Model 2 introduces a graduated framework built on top of the original F_Score (the A_Score), which scores each signal based on how strongly the firm has performed relative to its peers for that year. Every year, June 30 is taken as the portfolio formation date and is held for one year.

The original F_Score works with stocks scoring F_Score ≥ 7, outperforming the NIFTY 100 benchmark, and this effect is further strengthened when financial sector firms are excluded. The A_Score, however, does not perform better than the F_Score. The conclusion is that for NSE 100 investors, the F_Score, when applied to a long-only portfolio, is a simple, effective and well-performing strategy. Added complexity does not add value in this universe.

Keywords: *Piotroski F_Score, Value Investing, Fundamental Analysis, NSE 100, Relative Weighted Scoring, Book-to-Market Ratio, Indian equity market, large-cap stocks.*

Table of Contents

List of Tables

CHAPTER 1: INTRODUCTION

1.1. Background of the Study

Value Investing, an investment strategy of buying stocks that are undervalued relative to their fundamental worth, has a rich academic heritage. *Value premium* is one of the most persistent anomalies in finance, identified by high book-to-market stocks earning superior returns, as documented by (Fama & French, 1992). However, not all cheap stocks are cheap for a good number of reasons in their own right; many are priced low precisely because their financial condition is deteriorating. The challenge for the value investor is to distinguish genuinely undervalued firms from *value traps*.

(Piotroski, 2000) landmark paper addressed this challenge directly. Working with a large cross-section of US high-BM firms over 1976-1996, Piotroski designed a nine-signal composite score – the F_Score – that aggregates binary indicators of profitability, leverage, liquidity and operating efficiency. He showed that buying <u>financially strong value firms</u> and shorting financially weak ones generated a 23% annual return over his sample period, with the strategy robust across time and market conditions. The mean return to a high-BM investor could be increased by at least 7% annually by focusing on high F_Score firms.

Ever since his publication became well-spoken of in academic research, the F_Score has been tested and tried across various markets, at various time periods and intervals. Evidence from India is of particular interest. The Indian equity market, with its distinctive institutional features – including a large and growing retail investor base, concentrated ownership structures, mixed analyst coverage, and periodic liquidity constraints – may respond differently to fundamental signals compared to the US market from which the F_Score was originally developed. (Tripathy & Pani, 2017) tested the F_Score on NSE Top 500 companies and confirmed that high F_Score firms in India generate superior contemporaneous and future stock performance, though the relationship is strongest for future valuation multiples rather than raw returns.

1.2. Industry Context

The NSE 100 represents India's most liquid and institutionally traded large-cap equity index, and it is the most practically relevant for investors who benchmark against NIFTY 100. It exhibits structural characteristics – high analyst coverage, index-driven quality filtering and institutional investors' concentration – indicating that accounting-based signals could carry less predictive power.

Piotroski's universe contained no financial sector firms, but this was not a deliberate methodological exclusion. It was a consequence of his universe construction: banks, insurance companies, and other financial intermediaries typically carry low book-to-market ratios because their balance sheets consist primarily of financial assets marked close to market values. As a result, they did not qualify under his high BM screening criterion. Piotroski's nine signals were therefore implicitly calibrated for non-financial operating companies, not by design but by the composition of the universe his BM produced.

1.3. Problem Statement

The main limitation of the F_Score lies in its binary scoring nature. All nine individual signals have equal weights – every metric gets a score of one if the condition is satisfied, regardless of whether a firm's metric improved marginally or drastically. This, in essence, means that the F_Score cannot distinguish between firms that have performed much better than a firm that just went up with the averages. Modifications to the F_Score framework had been explored by (Sharma et al., 2021) to increase its discriminative power, and recognises that the degree of improvement on each dimension of the F_Score may carry more weightage over and above the binary score that indicates the direction.

1.4. Scope of the Study

This study looks at companies that were part of the NSE 100 in each of the 12 years (July 2013 to June 2025). NSE 100 was selected as the initial universe as it represents the country's most liquid and institutionally traded large-cap equity index. (Piotroski, 2000) supports the fact that the F_Score investing strategy is effective across all market cap segments. This makes NSE 100 a well-justified and relevant setting in which the F_Score is replicated and extended.

1.5. Research Gap

The literature review identifies two distinct gaps that makes the current study relevant. The first gap is geographical and universe specific: F_Score has been validated across global emerging markets and the broader NSE Top 500 universe; no prior study has tested the framework specifically within the NSE 100. This matters because the NSE 100 exhibits structural characteristics – high analyst coverage, institutional ownership and index driven quality filtering.

The second gap addresses the methodological aspect: (Sharma et al., 2021) did propose that modifications to F_Score's equal weighting can improve performance and (Tikkanen & Äijö, 2018) proposed that graduated measures should carry more information than binary ones. This study is the first to systematically explore this aspect, testing a graduated percentile-based version of the F_Score with specific reference to Indian equities. By running two models on identical samples and subsamples, this study aims to provide a clean comparison of the two methodologies, thereby contributing to the existing literature on both Indian equity markets and the design of accounting based investment screens.

1.6. Research Objectives

The primary objective of this study is to check whether a graduated F_Score (Model 2, A_Score) can increase the mean annual portfolio return achieved by an NSE 100 value investor, as compared to the original binary scoring model, while keeping a passive NIFTY 100 benchmark investment.

The current study contributes to several existing literatures. First, it applies the F_Score value investing strategy specifically to the NSE 100. Despite being strongly effective in mid-cap and small cap segments, (Piotroski, 2000) demonstrated that the F_Score strategy is effective across all market capitalisation segments. Therefore, this study provided evidence relevant to fund managers and investors who use NIFTY 100 as the benchmark index.

Secondly, a graduated scoring methodology, A_Score, is introduced that is an extension of the F_Score, which incorporates the magnitude of financial improvement in each signal of the F_Score, not merely its direction. To the best of our knowledge, this is the first study to systematically test the A_Score in the Indian large cap context. Studies in the past have looked at changing the composition of the metric in F_Score, but none have addressed the magnitude of improvement in the F_Score metrics.

Thirdly, this present study tries to provide a comparison across four distinct analytical subsamples – full sample and non-financial samples (subsample) – that allows clear attribution to the scoring methodology versus sample composition effects.

1.7. Research Questions

This present study asks this specific question:

RQ1: Does the Piotroski F_Score work well in India's NSE 100 universe, generating superior returns for high scoring firms?

RQ2: Does a graduated Piotroski F_Score – in which a firm is rewarded between 0-3 points on each signal based on its percentile rank within each year's universe – can generate portfolio returns that are higher than the returns when applied to the same NSE 100 universe?

CHAPTER 2: LITERATURE REVIEW

2.1. Theoretical Background

The study draws on multiple interconnected theoretical frameworks: Efficient Market Hypothesis, Value Premium and its mechanisms, fundamental analysis and signal-based investing, book to market ratio (BM Ratio) as a valuation construct. Each theory provides a justification layer for the research design and helps situate the study's empirical findings within the broader finance literature.

2.1.1. Efficient Market Hypothesis

In the finding of the EMH theory by (Fama, 1970), the author asserts that all available information is always fully incorporated into asset prices at any given time, which makes it impossible to earn abnormal return on systematic and sustained basis. Three forms of classifications exist under EMH, which are based on a set of information that is assumed to be reflected in prices: weak form (only historical price data), semi-strong form (all publicly available information) and the strong form (all information, including private sources). For the current study the semi-strong is most relevant. The findings of this study will provide us with a view that if NSE 100 stock prices fully reflect all publicly available accounting information, then the signals as defined under the F_Score framework should carry no predictive power over future returns.

The null hypothesis is based out of this proposition of EMH theory against which empirical results are evaluated (discussed further in the document). The documented ability of accounting-based signals to predict future returns – confirmed by (Piotroski, 2000) in the US and by (Tripathy & Pani, 2017) in India – constitutes evidence against strong form semi-strong efficiency in these markets.

(Hyde, 2013), in his study had demonstrated that even among institutionally prominent and high coverage stocks, F_Score still has predictive power. This suggests that markets underreact to gradual improvements in financial fundamentals even if it is visible to all market participants. This finding provides theoretical grounding for the expectation that the strategy should retain predictive power within the NSE 100 despite its high analyst coverage and institutional ownership.

2.1.2. The Value Premium: Risk-based and Behavioural Explanations

The empirical relationship between book-to-market (BM) ratios and stock returns has been extensively documented. Firms with high BM ratios – the *value stocks* – consistently outperform low-BM growth stocks over long horizons. (Fama & French, 1992) had documented this relationship empirically, showing that the BM ratio is one of the strongest predictors of cross-sectional stock returns and (Fama & French, 1993) formalised this relationship within their three-factor model, proposing that the value premium reflects either compensation for a systematic factor or persistent mispricing. (La Porta et al., 1997) provided behavioural explanations, arguing that investors extrapolate the recent performance into stock prices too far into the future, causing the momentum stocks (growth stocks) to be overpriced relative to their counterparts of value stocks. Regardless of this mechanism as well, the *value premium* has been one of the most replicated findings in empirical asset pricing.

In the Indian context, value premium is well established. High BM firms with Indian equity markets have historically generated above benchmark returns, a finding consistent with global evidence. However, within high BM firms, return differences are substantial: many high BM firms are cheap precisely because their financial condition is deteriorating, and as such, investors tend to extrapolate these poor earnings into the future, causing them to be underpriced. This observation motivates the use of fundamental analysis to separate financially viable value firms from distressed ones.

International robustness of the value premium provides an important justification for the present study. (Fama & French, 1998) extended the BM-return relationship to 13 major international markets over the period of 1975-1995, finding that value stocks outperformed growth stocks in 12 of those 13 selected markets. This global evidence confirms that the value premium is not specific to only US markets, and this provides a strong backing for applying this value-stock screening strategy, such as the score used in this study, in the Indian large-cap context. The persistence of the value premium across markets with varying levels of efficiency and institutional development suggests that the principle of value stock outperformance is sufficiently robust to support fundamental analysis-based investment strategies in the NSE 100 universe.

2.1.3. Fundamental Analysis and Financial Statement Information

Fundamental Analysis rests on the fact that a firm's intrinsic value is determined by its capacity to generate sustainable earnings and cash flows, and that this value is estimable from publicly available financial statements, which provide the structured information on profitability, operating efficiency, liquidity and solvency – the four dimensions that (Piotroski, 2000) systematised into his nine-signal scoring framework. Accounting ratios can transform raw financial data into standardised information that can be cross-sectionally compared across different stocks within a given valuation cohort, which allows investors to distinguish between improving firms and deteriorating ones.

The rationale for fundamental analysis, both theoretical and empirical, is the recognition of the fact that financial markets do not fully and immediately reflect changes in financial condition. This is particularly true for value stocks, as the investor base for these types of stocks is inherently sceptical, and negative sentiment from the past lingers even as the underlying fundamentals improve. This lag between improvement and stock price reaction represents a predictable return for the investor base, as they respond to changes signalled by financial statements. This represents the underlying theoretical rationale for the use of the F_Score as presented here.

2.1.4. Signal-Based investing and F_Score as a Multi-signal framework

Signal-based investment frameworks propose that individual accounting signals — each capturing a distinct dimension of financial health — can be combined into a composite indicator that is more powerful than any single signal in isolation. A firm may improve on cash flow generation, but simultaneously it may have increased leverage, which produces a mixed fundamental picture. This obstacle can be clearly tackled by aggregating the various dimensions of a company – profitability, leverage, liquidity and operating efficiency – thereby a composite score encompassing these can accurately measure a firm's performance as compared to basing the opinion on any single signal. F_Score provides an aggregate measure of financial quality by summing the nine signals across the 4 dimensions and filters out 'noise' inherent in individual ratios.

F_Score serves as the second stage of filtering in the framework. The first screening of the BM ratio filters in stock that are genuinely undervalued, and the F_Score then separates firms with improving fundamentals from those with deteriorating ones (value traps). This combination drives the structural design of both the models involved in his study.

2.2. Piotroski's F_Score Framework

A nine point composite scoring was proposed by (Piotroski, 2000) to differentiate between winners and losers within the value universe. The F_Score consolidates binary signals across three dimensions: Profitability – ROA, Cash flow from Operations, Changes in ROA, Earnings Quality through the Accruals Measure; Leverage and Liquidity – Change in Long Term Leverage, Change in Current Ratio, Change in Equity Issued; Operating Efficiency – Change in Gross Margin, Change in Asset Turnover. Each signal assigns one point, resulting in a score ranging from 0 to 9.

Piotroski's key empirical findings are: (a) Strong Firms (F_Score of 8 & 9) earned approximately 13.4% more than Weak firms (F_Score of 0 & 1) on a market adjusted basis; (b) a long-short hedge strategy buying Strong and shorting Weak firms generated a 23% annual return over 1976-1996; (c) the strategy was robust across sub-periods and market conditions; and (d) the benefits to financial statement analysis were concentrated among small-cap and low analyst following firms, suggesting that information inefficiency was a key driver.

2.3. International Replication Evidence

The context of this study is further supported by emerging markets evidence. (Hyde, 2013) tested the same F_Score across MSCI Emerging markets and found a significant return premium across F_Score stocks, after accounting for risk factors. Crucially enough, Hyde's Analysis found that the strategy was most effective among well-covered, institutionally prominent stocks rather than neglected micro-caps, suggesting that the usage of the F_Score genuinely captures fundamental information that is incorporated into prices even for high-visibility firms, attributing this to confirmation bias among investors who respond to fundamental improvements only when they are already in the public spotlight.

(Tikkanen & Äijö, 2018) examined whether F_Score improves the performance of different value investment strategies in European equity markets, finding that F_Score screening adds incremental return across multiple value definitions (BM, earnings yield, dividend yield and cash flow yield), confirming that the strategy's effectiveness is not contingent on a single valuation metric.

(Walkshäusl, 2020) extended this evidence to a global multi-country sample, documenting consistent F_Score return internationally after risk adjustment. Taken together, this body of evidence establishes that the F_Score is a robust cross-market strategy, providing a strong prior expectation that it should function in the Indian large-cap context.

2.4. F_Score Evidence from India

The most directly relevant study for the current study is by (Tripathy & Pani, 2017), wherein the F_Score investing framework was applied to the NSE 500 companies over multiple years, looking at panel data regression models to estimate the relationship between F_Score and three performance measures: Stock Returns, ROE, Market to Book Value. Their key findings were that high-BM firms (value stocks) with high F_Scores shifted the distribution of both contemporaneous and future stock performance in favour of investors. They concluded that the F_Score provides useful incremental information to Indian equity investors, even in the context of a broader, more liquid universe than Piotroski's original sample.

The study by (Tripathy & Pani, 2017) used NSE Top 500 rather than NSE 100, which means their sample includes a much larger universe involving mid-cap and small-cap segments as well. The present study deliberately focuses on the NSE 100 specifically. (Piotroski, 2000) demonstrated that the F_Score strategy is not size dependent: it produced positive return differentials across small, medium and large market caps in the US market, although the predictive power of the F_Score was concentrated more among small-cap firms with low analyst coverage.

The NSE 100, therefore, represents a valid and practically relevant test for the strategy and grounds the expectation that the strategy should retain predictive power despite high analyst coverage.

2.5. Modifications and Extensions to the F_Score

Several researchers proposed modifications to Piotroski's original framework to improve its predictive power. (Sharma et al., 2021) examined the Piotroski score's constituent parameters and proposed adjustments designed to achieve a better Sharpe Ratio at lower portfolio beta. Their work identified shortcomings in the equal-weighted assumption, specifically the inability of the binary scoring to capture the magnitude of financial improvement. By adjusting the scoring weights to reflect the economic importance of each signal, they were able to generate portfolios that increased their risk-adjusted performance in their test sample.

The literature on quantitative factor investing suggests that continuous or graduated measures of factor exposure tend to provide more information than simple binary classifications. The

Sharpe ratio maximisation framework implies that the weight assigned to a stock in a portfolio should be proportional to its expected excess return conditional on all available information – a framework that points to the usage of graduated rather than binary scores. Model 2 is built on this motivation.

CHAPTER 3: CONCEPTUAL FRAMEWORK

3.1. Variable Operationalisation

Table 1 below defines the dependent, independent, moderating and benchmark variables used in the current study. The study employs a quantitative, deductive research design, using financial statements and market price data to test the hypotheses. The dependent variable is the mean annual equal weighted portfolio return; the independent variables are the F_Score Group (Model 1) and the A_Score group (Model 2), each capturing the level of financial strength of a value stock each year. Financial sector membership moderates both relationships by partitioning the sample into full and non-financial subsamples. The causal logic is that financial statement signals (IVs) predict future portfolio returns (DV), with the direction of causality running from accounting fundamentals to stock price performance.

Table 1: Variable Definition

Variable	Role	Definition	Source
Annual Portfolio Return	Dependent Variable	Equal weighted mean annual stock return within each F_Score / A_Score group, measured from June 30 of year t to June 30 of year t+1 using adjusted closing prices. $$R = \frac{Price_{t+1} - Price_t}{Price_t}$$	Capitaline Adjusted Closing Price (Base as NSE adjusted Closing Prices)
F_Score Group (Model 1)	Independent Variable	Categorical: Strong (F≥7), Neutral (4-6); Weak (F≤ 3). Derived from the sum of nine binary accounting signals (0-9 scale)	Capitaline Financial Statements
A_Score Group (Model 2)	Independent Variable	Categorical: Strong (A≥15), Neutral (A_Score = 9-14), Weak (A < 9). Derived from the sum of nine graduated signals (0-27 scale), where each signal is awarded 0-3 based on year-specific percentile rank.	Capitaline Financial Statements
NIFTY 100 benchmark return	Benchmark Reference	Annual Total Returns of the NIFTY 100 index from June 30 of year t to June 30 of year (t+1). Used as the passive investment benchmark against which portfolio returns are compared. Serves as a practical performance reference for H1.	Adjusted Closing price from Capitaline Database

BM Ratio	Screening Criterion	BM = Book Value of Equity / Market Capitalisation at fiscal year-end. Used to identify value stocks (top 30% per year). This defines the eligible universe for F_Score and A_Score analysis.	Capitaline (Book Values & Market Capitalisation)

Note: The study design is cross-sectional within years and time series across 12 annual portfolio formation periods (2013-2024). The unit of analysis is the firm-year observation. Statistical inferences are based on the time series of 12 annual group mean returns, consistent with Piotroski's (2000) methodology. No multivariate regression framework is applied; the study uses single sample t-tests on portfolio return time series, the standard approach in portfolio-sort-based empirical asset pricing research.

3.2. Hypothesis Development

Two related yet distinct questions form the motivations for this study. The first replicates (Piotroski, 2000) core proposition in the Indian large-cap context: can accounting-based signals separate fundamentally strong value stocks from weak ones in a way that translates into superior returns? The second extends its proposition by asking whether replicating the binary scoring scheme with a graded, continuous measure of financial strength can further improve discriminative power. These motivations are formalised in the following two hypotheses:

H1 (Replication): *Within the NSE 100 value stocks universe, financially strong firms (F_Score ≥ 7) generate significantly higher mean annual returns than financially weak firms (F_Score ≤ 3).*

H2 (Extension): *The graduated A_Score (Model 2) produces a larger strong-weak return spread than the binary F_Score (Model 1) within the same NSE 100 value stock universe.*

H1 is a replication hypothesis, wherein this study tests whether the strategy holds in a well-defined, practically investable large-cap universe where the market is relatively efficient, and the proportion of distressed firms is structurally smaller than in small-cap universes. H2 is the original contribution of this study, motivated by the theoretical argument that a continuous measure of financial health should carry more information than a binary one.

CHAPTER 4: RESEARCH METHODOLOGY

4.1. Research Design

The research design is quantitative and exploratory: secondary sources of data from publicly available financial statements and market price records are collected. The analytical methodology that is followed is portfolio-sorted, which is the standard in empirical asset pricing studies. Firm-year observations become the unit of analysis for this study, within an annually rebalanced long-only portfolio framework.

4.2. Data Sources

This study universe consists of all companies that were constituents of the NSE 100 index at any point during the period from July 2013 to June 2025. Index membership lists were sourced for each year from NSE India's published constituent disclosures, covering year-by-year compositions from 2013 to 2024. In total, 189 unique companies appear in the dataset across the 12 portfolio formation years.

The total firm-years that had to form part of the initial universe are 1202. However, the realised dataset contains 1,049 firm-year observations, representing a shortfall of 153 observations. This attrition arises from two distinct sources: (a) data for 125 out of 153 firm-year observations that appeared in the NSE 100 index during the study window were absent from the Capitaline database and could not be included in any year. This group includes companies that were newly listed after the study period and therefore had no prior financial history in Capitaline, companies that were delisted or underwent corporate restructuring before data collection and companies for which Capitaline data were either unavailable or inaccessible at the time of collection. (b) A further 28 firm-years represent isolated gaps for companies that are otherwise present in the dataset: data was obtainable for most years. These gaps reflect year-specific unavailability in Capitaline rather than structural exclusion. The resulting 1,049 firm-years cover 84 to 93 companies per year across the 12-year sample period.

The parameters that form part of the nine-point scoring metric, as per (Piotroski, 2000) – Net Income, Total Assets, Cash Flow from Operations, Long Term Debt, Current Assets, Current Liabilities, Equity Shares Outstanding, Gross Profit and Net Sales – were sourced from the Capitaline database for all eligible companies. Financial data for F_Score computation was sourced from Capitaline, and adjusted closing price was also gathered from Capitaline, with

NSE, with returns calculated as the percentage change in adjusted price from June 30 (portfolio formation date) of year t to June 30 of year t+1. NIFTY 100 index return data was collected from the official index history provided by the NSE website.

The adjusted closing prices have been taken into consideration for the calculation of stock splits, bonus issues, and dividends paid during the period, as this ensures that the returns calculated are the actual economic returns for the buy-and-hold investor. The portfolio formation date of June 30 ensures that the annual financial statements for the companies (most of which have their financial year-end in March) have been publicly disclosed before the portfolios were formed, thus satisfying the look-ahead bias constraint. A post-hoc audit of all financial data confirmed that every firm year in the final sample has a March fiscal year-end and that the returns for each firm year observation are measured from June 30 immediately following the fiscal year-end, so no financial data enters the portfolio before it is publicly available. A post-hoc audit reveals that the final sample does not include any observation with a fiscal year end other than March.

4.3. Sampling Method and Sample Size

Following (Piotroski, 2000) value stocks are identified as those in the top 30% of BM ratios within the NSE 100 universe in each year. The BM ratio is computed as:

BM Ratio = Book Value of Equity / Market Capitalisation

where the book value is derived from the most recently available audited financial statements before the portfolio formation date, and the market capitalisation is calculated as the shares outstanding times the adjusted closing price as of the fiscal year-end. An eligibility criterion is imposed such that if the company's fiscal year-end is on or before June 30, the current year financial statements are used, while if the company's fiscal year is after the portfolio formation date of June 30, the prior year financial statements are used to ensure only publicly available information is used in the ranking.

Companies are ranked based on the BM ratio calculation yearly, with the top 30% of the companies identified as value stocks, which translates to 25 to 28 companies per year, depending on the universe size, for both F_Score and A_Score calculations. Over the full 12-year period, this produces 316 value stock firm year observations over the full sample and 251 from the non-financial subsample. Table 2 summarises the year-by-year composition of the value stock universe. Within this universe, two additional firm-year observations were excluded from the F_Score Analysis in 2013, by reason of lack of prior year (t-1) financial data in the Capitaline database, making it impossible to compute the change-based signals – ΔRoA, ΔLeverage, ΔLiquidity, ΔGross Margin, and ΔAsset Turnover. This is consistent with the standard practice in F_Score replication studies, which require at least two consecutive years of financial data per firm-year observation. The final analytical sample is therefore 316 (+2 dropped, due to lack of data) firm years for the full sample and 251 (+3 dropped, due to lack of data) firm years for the non-financial subsample.

Table 2: Value Stock Universe: Year-by-Year Comparison

Year	Full Sample Universe	Full Sample Value Stocks	Subsample	Subsample Value Stocks
2013	84	25 (-2)	63	19 (-1)
2014	85	26	63	19
2015	86	26	66	20
2016	88	27	73	22
2017	89	27	73	22
2018	89	27	70	21
2019	85	26	68	21
2020	85	26	67	20
2021	88	27	74	22
2022	87	26	71	22
2023	90	27	76	23 (-2)
2024	93	28	76	23
Total	**1049**	**318 (-2)**	**840**	**254 (-3)**

Note: Non-financial value stocks exclude banks (both public and private sector), financial institutions, NBFCs, finance companies, general insurance, life insurance, holding companies, other financial services, AMCs, and housing finance corporations. The exclusion covers 64 firm-years from the full sample of 318. (+/- n) represents the number of firm-year observation against a particular year that were dropped due to lack of data for that specific year

Descriptive statistics for annual portfolio return across all four analyses – including mean returns, standard deviations, and return ranges for each group – are presented in Section 5, alongside the corresponding hypothesis test results.

4.4. Scoring Methodology

4.4.1. Model 1: Binary F_Score

Model 1 replicates (Piotroski, 2000) original nine-binary F_Score. Each of the nine signals is assigned a value of 1 if the underlying financial condition meets the criterion, and 0 otherwise. The F_Score is the sum of all nine binary signals, ranging from 0 to 9. Table 2 presents the nine signals, their categories and the scoring criteria.

Table 3: Piotroski F_Score Signals

Signal	Category	Formula/Criterion	Score = 1, if
F1	Profitability	RoA_t = NI_t / TA_t	RoA_t > 0
F2	Profitability	CFO Ratio = CFO_t/TA_t	CFO Ratio > 0
F3	Profitability	ΔRoA = RoA_t-RoA_t1	ΔRoA > 0
F4	Profitability	Accrual = F2-F1	Accrual > 0 (i.e., F2 > F1)
F5	Leverage & Solvency	ΔLEV = (LTD_t / TA_t)-(LTD_t1 / TA_t1)	ΔLEV < 0
F6	Leverage & Solvency	ΔLiquidity = CR_t - CR_t1	ΔLiquidity > 0
F7	Leverage & Solvency	Equity Offering = Shares Outstanding_t - Shares Outstanding_t1	No New Equity Issued
F8	Operating Efficiency	ΔMargin = (GP_t / NS_t) - (GP_t1 / NS_t1)	ΔMargin > 0
F9	Operating Efficiency	ΔATR = (NS_t / TA_t) - (NS_t1 / TA_t1)	ΔATR > 0

Note: t represents data as at immediately completed fiscal year, and t1 represents data as of t-1. CFO = Cash Flow from Operations, NI = Net Income; TA = Total Assets; LTD = Long Term Debt; CR = Current Ratio; GP = Gross Profit; NS = Net Sales; RoA = Return on Assets. Signals F1 through F4 relate to profitability, F5 through F7 to leverage and liquidity; F8 and F9 to operating efficiency. All inputs derived from the most recently available audited annual financial statements as of June 30 of the portfolio formation year.

Portfolio grouping in Model 1 follows Piotroski's convention, adapted to the size of annual sub-samples. Value stocks are classified as Strong (F_Score ≥ 7), Neutral (4 ≤ F_Score ≤ 6) and Weak (F_Score ≤ 3). These thresholds are consistent with the spirit of Piotroski's original classification and represent scores of 77.8%, 44.4% and ≤ 33.3% of the maximum possible score, respectively. The thresholds are constant across all years, ensuring full cross-year comparability.

4.4.2. Model 2: Graduated A_Score

Model 2 introduces a graduated A_Score that retains the nine-signal structure of Model 1 but replaces the binary scoring with a 0-3 scale for each signal. The motivation is to capture the magnitude of financial improvement, not just the direction. A firm that improved its performance drastically in a metric should be rewarded more than a firm that improved it meagerly.

The graduated weighted scale is implemented through year-specific percentile thresholds. For each signal and each year, the 33rd and 67th percentiles of the signal's distribution across the full value stock are computed. A firm receives:

- 3 points: if the signal value falls above the 67th percentile (P67).
- 2 points: if the signal value falls between the 33rd and the 67th percentile (P33 and P67).
- 1 point: if the signal value falls in the 33rd percentile, but still satisfies the directional test (i.e., directionally positive – the performance improved).
- 0 points: if the signal fails the directional test entirely (directionally negative – the performance deteriorated).

  The directional test differs by signal type. For level signals of A1 and A2, a value below zero is given a score of 0. For change signals A3, A4, A6, A8, A9, a negative change gives a score of 0. For A5, any increase scores 0, and the percentile thresholds are applied to the absolute magnitude of the decrease. For A7, any net increase in shares outstanding scores 0; the percentile thresholds are applied to the magnitude of the share reduction.

The year-specific percentile thresholds are computed separately for the full sample and for the non-financial sub-sample, ensuring that the scoring benchmark is appropriate for the comparison group. The A_Score is the sum of all nine graduated signals, ranging from 0 to 27. The table below summarises the nine graduated signals.

Table 4: Graduated A_Score Signal Definitions

Signal	Category	Graduation Criterion	Scoring (0-3)
A1	Profitability	RoA_t level	Negative = 0; 1 = Positive but ≤ P33; 2 = between P33 and P67; 3 = ≥ P67
A2	Profitability	CFO Ratio	Negative = 0; 1 = Positive but ≤ P33; 2 = between P33 and P67; 3 = ≥ P67
A3	Profitability	ΔRoA percentile rank within the year	0 = wrong direction; 1 = positive direction but < P33; 2 = between P33 and P67; 3 = ≥ P67
A4	Profitability	Accrual (A2-A1) percentile rank within year	0 = negative (accruals > cash earnings); 1 = positive but < P33; 2 = between P33 and P67; 3 = ≥ P67.
A5	Leverage & Solvency	ΔLEV percentile rank within year	0 = leverage increased; 1 = decreased but < P33; 2 = between P33 and P67; 3 = ≥ P67
A6	Leverage & Solvency	ΔLiquidity percentile rank within the year	0 = wrong direction; 1 = positive direction but < P33; 2 = between P33 and P67; 3 ≥ P67
A7	Leverage & Solvency	ΔShares % percentile rank within the year	0 = shares increased; 1 = no increase but < P33; 2 = between P33 and P67; 3 = ≥ P67.
A8	Operating Efficiency	ΔMargin percentile rank within the year	0 = wrong direction; 1 = positive direction but < P33; 2 = between P33 and P67; 3 ≥ P67
A9	Operating Efficiency	ΔATR percentile rank within the year	0 = negative direction; 1 = positive direction but < P33; 2 = between P33 and P67; 3 ≥ P67.

Note: Year-specific thresholds (P33 and P67) are computed separately for each signal within each portfolio year, based on the distribution of the relevant financial metric. Full percentile thresholds are attached in the appendix.

Portfolio grouping in Model 2 was an absolute threshold on the A_Score.

- Strong: A_Score ≥ 15
- Neutral: 9 ≤ A_Score ≤ 14
- Weak: A_Score ≤ 9

The determination of the thresholds involved three sequential steps: the first step was to establish that the group should use absolute rather than relative thresholds. An initial specification group stocks by their percentile rank based on the A_Score within each year: the top 33% was designated as Strong, the middle 34% Neutral, and the bottom 33% Weak. This approach was rejected because it produces results that are not comparable with Model 1's absolute F_Score thresholds. Under a relative approach, a stock scoring 14 in a weak year would be classified as strong, while the same score in a weak year would push the stock into Neutral. Direct comparison of the two models requires that both apply absolute thresholds every year. The second step was to apply equal-interval absolute thresholds by dividing the score range of 0-27 into 3 groups: Weak: 0-8; Neutral: 9-17; Strong: 18-26. Under this specification, the Strong group was too small across most years to support meaningful statistical analysis.

The third step was to lower the threshold of what classifies a score as Strong iteratively, until the resulting group held a sample that was comparable to Model 1's portfolio of stocks. The threshold of A Score ≥ 15 was arrived at through this iterative process, producing a total of 50 Strong firm-years across the full sample compared to Model 1's 88 firm-years for the same group.

4.5. Portfolio Construction and Return Measurement

For both models, portfolios were formed on June 30 of each year and held for exactly one year until June 30 of the following year. Within each group, returns are computed as the equal-weighted average of individual stock returns during the holding period. The equal-weighted assumption is consistent with (Piotroski, 2000) methodology. The annual portfolio return is the mean of the returns of stocks in that group for that year.

With respect to the hedge portfolio, following the methodology of (Piotroski, 2000) the return is computed as the arithmetic difference between the returns of the Strong Portfolio and Weak Portfolio in the same year. This is an academic construct that is being used to evaluate the predictive power of the F_Score, which essentially removes the common factors that could affect all high-BM stocks equally, such as general market movements and momentum effects, thereby isolating the contribution of financial statement signals alone. In years where either of the groups has no members, the hedge spread is undefined and is simply excluded from the mean calculation.

4.6. Treatment of Financial Sector Companies

Financial sector firms – banks, NBFCs, Insurance companies and holding companies – present a methodological challenge. Piotroski's nine signals were designed for operating companies: leverage for a bank is a business model characteristic rather than a risk indicator, and the CFO is driven by loan disbursement rather than operating performance. In Piotroski's original study, financial firms were absent from the high BM-firms by construction, as US financial firms carried low BM ratios and did not pass the screening filter.

In the NSE 100, the situation is different. Financial firms constitute approximately 38% of the value stock universe. Indian financial sector companies have carried depressed market valuations relative to book value for much of the study period, causing them to rank highly on the BM ratio and qualify themselves into the top 30% criteria.

The study, therefore, includes all value stocks, including financial firms, as this reflects the full NSE 100 investment universe, and it would present itself to a practising investor. It was also taken into consideration that Piotroski's signals were designed for operating companies, and given the fact that financial firms constitute such a large, distinctive share of the value stock universe, a pre-planned sensitivity analysis was conducted on a non-financial subsample (n=251 firm-years). The rationale for this is to check whether the inclusion of financial firms materially affects the results.

4.7. Statistical Tests

Statistical significance is assessed using single-sample t-tests on the time series of annual portfolio returns, with the null hypothesis that the true mean return equals zero.

The same t-test is applied to the time series of Strong-Weak annual spreads to test whether the spread is significantly different from zero. Since the annual return measurement period is 12 years, the degrees of freedom are either 10 or 11, depending on the number of years in which the relevant group has members. The reported p-values are two-tailed, and significance is indicated at the 5% level (**) and 10% level (*).

The current study does not incorporate any formal multi-factor risk adjustment, consistent with (Piotroski, 2000). To partially account for the value premium that generally applies to high BM stocks, the study also benchmarks the Strong portfolio against the NIFTY 100 index return (a proxy for market factor). The strong portfolio (F_Score $\geq$ 7) outperforms the benchmark, which suggests that the score adds incremental return beyond holding value stocks within the NSE 100 universe. The primary metric used for hypothesis testing is the mean annual return of each portfolio group and the associated t-statistic.

CHAPTER 5: DATA ANALYSIS & FINDINGS

5.1. Descriptive Statistics

Table 5 reports the descriptive statistics for annual equal-weighted portfolio returns across all four analyses. These statistics summarise the distribution of annual group means returns over the 12-year sample period and provide the baseline for the hypothesis tests that follow

Table 5: Descriptive Statistics: Annual Equal-Weighted Portfolio Return

Portfolio Group	N (Years)	Mean Return	Std. Dev.	Min	Median	Max
Panel A: Model 1: Full Sample (n=316 firm years)						
Strong (F ≥ 7)	12	25.18%	34.69%	(23.28%)	29.04%	83.92%
Neutral (4-6)	12	19.18%	31.18%	(24.21%)	13.68%	77.33%
Weak (F ≤ 3)	11	14.10%	34.25%	(37.00%)	12.94%	65.75%
NIFTY 100 Benchmark	12	14.63%	—	—	—	—
Panel B: Model 1: Subsample (n=251 firm years).						
Strong (F ≥ 7)	12	29.29%	44.57%	(19.67%)	15.95%	107.58%
Neutral (4-6)	12	19.74%	37.91%	(22.67%)	5.87%	96.26%
Weak (F ≤ 3)	12	14.52%	28.44%	(30.21%)	11.90%	61.06%
Panel C: Model 2: Full Sample (n=316 firm years)						
Strong (F ≥ 15)	11	16.09%	28.27%	(19.83%)	20.31%	63.62%
Neutral (9-14)	12	19.15%	28.80%	(25.28%)	9.65%	63.97%
Weak (F < 9)	12	22.69%	34.51%	(26.19%)	15.96%	96.22%
Panel D: Model 2: Subsample (n=251 firm years).						
Strong (F ≥ 15)	12	27.22%	39.58%	(23.99%)	31.59%	107.58%
Neutral (9-14)	12	19.77%	37.35%	(26.72%)	1.74%	86.48%
Weak (F < 9)	12	22.05%	34.50%	(21.34%)	11.00%	101.73%

Note: Each observation is the equally weighted average of all stocks in that group for a given year (June 30 of time t to June 30 of time t+1). N = number of years with at least one member in the group (Weak group had 0 stocks for the year 2023 – hence N = 11 in this case). Statistics *are computed on the time series of annual group means, consistent with t-tests reported in Section 4. Std. Dev. = Standard Deviation of annual mean returns. Figures in () represent negative returns.*

5.2. Results by Model

5.2.1. Model 1

Table 6 provides the distribution of stocks based on F_Score values across the value stocks under Model 1 for both Full sample and non-financial subsample.

Table 6: Distribution of F_Score values across the Value Stock Universe

F_Score	Group	N (Full Sample)	% of Full Sample	N (Subsample)	% of subsample
0	Weak	1	0.32%	0	0.00%
1	Weak	3	0.95%	1	0.40%
2	Weak	6	1.90%	6	2.39%
3	Weak	30	9.49%	23	9.16%
4	Neutral	71	22.47%	41	16.33%
5	Neutral	67	21.20%	45	17.93%
6	Neutral	50	15.82%	50	19.92%
7	Strong	58	18.35%	50	19.92%
8	Strong	27	8.54%	29	11.55%
9	Strong	3	0.95%	6	2.39%
Total		**316**	**100.00%**	**251**	**100.00%**

Note: F_Score values range from 0 (worst) to 9 (best). The distribution confirms the structural thinness of the Weak group in the NSE 100 universe. Non-financial subsample excludes all financial sector and holding company firm years (65 observations)

5.2.1.1. Model 1: Full Sample

Table 7 represents the year-by-year portfolio returns for Strong, Neutral and Weak groups under Model 1. The table also contains the average return over the study period earned in NIFTY 100, the value stock portfolio return (based on the BM criteria used in this study), the Strong-Weak hedge spread and the Strong-NIFTY 100 hedge spread.

Table 7: Model 1_Full Sample: Annual Portfolio Returns by F_Score Group (2014-2025)

Year	Strong	Neutral	Weak	NIFTY 100	Value	S-W	S-100
2014	48.03	54.67	65.73	31.98	56.50	(17.70)	16.05
2015	39.62	(12.52)	(7.42)	11.49	0.10	47.05	28.13
2016	27.48	(2.47)	(1.08)	(0.29)	1.15	28.57	27.78
2017	35.83	31.73	38.03	16.86	34.61	(2.21)	18.96
2018	(3.68)	11.60	(37.00)	11.77	3.01	33.32	(15.45)
2019	(1.98)	15.76	22.45	8.08	11.08	(24.43)	(10.07)
2020	(23.28)	(24.21)	(27.44)	(11.74)	24.30	4.16	(11.54)
2021	83.92	77.33	61.54	52.18	76.01	22.38	31.73
2022	(8.80)	(5.96)	(10.23)	(0.19)	7.59	1.43	(8.61)
2023	30.59	32.36	N/A	19.90	31.61	N/A	10.69
2024	79.31	53.97	37.54	30.93	56.23	41.77	48.38
2025	(4.84)	(2.12)	12.94	4.53	(0.94)	(17.78)	(9.36)
Mean	**25.18**	**19.18**	**14.10**	**14.63**	**19.79**	**10.60**	**10.56**
p-value	**0.029****	**0.057**	**0.202**	**---**	**---**	**0.195**	**0.112**

*Note: Figures (except p-value) are in %. Returns are computed as equal-weighted mean annual stock return within each group, measured from June 30 of year t to June 30 of year t+1. The "Year" label refers to the year in which the returns are measured. N/A indicates that no firm-year observation existed for that particular year. For calculating hedge returns and subsequent p-values, the year in which any portfolio contains N/A is removed. (** denotes statistical significance at 5% level – two-tailed t-test, null; mean = 0). Figures in () represent negative returns*

The results are consistent with Piotroski's original hypothesis. Across the 12-year sample, the mean annual return of the Strong portfolio (25.18%) is higher than that of the Neutral portfolio (19.18%), which in turn is higher than that of the Weak portfolio (14.10%). From a theoretical point of view, it is also true that the global mean return would follow this monotonic order. The NIFTY 100 benchmark return of 14.63%, when contextualised in this fashion, is also exceeded by both Strong and Neutral Portfolios.

The mean return of the strong portfolio is statistically significant at 5% level ($t = 2.52$, $p = 0.029$), which indicates that this strategy of holding financially strong stocks within the NSE 100 value stocks generates returns which are meaningfully different from zero. The other two portfolios' significance test is not undertaken, because they do not form part of the initial hypothesis of this study.

This thinning out of the Weak group is also a significant finding, and it is particular to the NSE 100. While Piotroski's distressed small-cap population had many financially weak firms, the NSE 100 comprises the 100 largest firms by free-float market capitalisation in India. In the NSE 100, the severely weakened firms are few and far between due to the large-cap nature of the index. Among those that are represented in the high BM group, many are cyclical heavyweights (steel, refining, commodities) that are experiencing a sector trough and perform well on the way out.

The hedge portfolio return (Strong-Weak spread) is a measure of the incremental predictive power of the F_Score after controlling for common factors that affect all high BM stocks. The reason is that since both Strong and Weak are from the same high BM sample in the same year, common factors like market returns, value premiums, and macroeconomic variables are controlled for in the spread between Strong and Weak. The spread between Strong and Weak, therefore, captures only the relevant information from the F_Score signals.

The hedge spread of the Strong-Weak averages out to 10.60% per year, which is economically meaningful, but this is not statistically significant at conventional levels of 5% and 10%. (p-value: 0.195, t = 1.36). Similarly, the Strong-NIFTY 100 spread provides a mean of 10.56% per year (t: 1.173, p-value: 0.112). This limited statistical significance reflects the property of the large-cap index universe: relatively stable financial profiles constitute the NSE 100. As a result, the Weak group is thin in certain years: in 2023, there were zero Weak Stocks, and in several other years, the maximum number of stocks in the Weak Portfolio was not more than 8.

Nevertheless, it is critical to note that such a figure should not be construed as the potential return on a tradable strategy. While shorting individual NSE 100 stocks on a sustained year-over-year basis is not without significant transaction costs, stock borrowing costs (or using stock futures at the individual contract level), and execution challenges that would greatly erode the potential return on such a strategy, the hedge return should be construed more along the lines of an academic measure of signal effectiveness.

The relevant and actionable result for investors is the Strong portfolio return of 25.18% on a year-over-year basis, which is more than double the NIFTY 100 benchmark return of 14.63% and does not involve shorting any position.

5.2.1.2. *Model 1: Non-Financial Subsample*

Table 8 presents the Model 1 results after excluding all financial sector firms and holding companies from the value stock universe, retaining 251 firm-year observations across 12 years.

Table 8: Model 1_Subsample: Annual Portfolio Returns by F_Score Group (2014-2025)

Year	Strong	Neutral	Weak	NIFTY 100	Value	S-W	S-100
2014	107.58	63.42	61.06	31.98	56.50	46.52	75.59
2015	5.43	(11.00)	20.67	11.49	0.10	(15.24)	(6.06)
2016	35.59	1.55	(20.95)	(0.29)	1.15	56.54	35.88
2017	35.83	18.41	25.10	16.86	34.61	10.72	18.96
2018	(7.64)	10.19	(1.52)	11.77	3.01	(6.12)	(19.42)
2019	(1.98)	(0.10)	2.45	8.08	11.08	(4.43)	(10.07)
2020	(19.67)	(22.67)	(30.21)	(11.74)	(24.30)	10.54	(7.93)
2021	90.67	96.26	41.46	52.18	76.01	49.20	38.48
2022	(8.74)	(5.49)	26.56	(0.19)	(7.59)	(35.30)	(8.55)
2023	26.47	15.06	(6.47)	19.90	31.61	32.94	6.57
2024	92.32	76.39	52.95	30.93	56.23	39.37	61.39
2025	(4.39)	(5.19)	3.12	4.53	(0.94)	(7.51)	(8.91)
Mean	**29.29**	**19.74**	**14.52**	**14.63**	**19.79**	**14.77**	**14.66**
p-value	**0.044****	**---**	**---**	**---**	**---**	**0.111**	**0.133**

*Note: Figures (except p-value) are in %. Excludes banks, NBFCs, financial institutions, insurance companies, holding companies and other companies which has nature of a financial company. All 12 years have data for all three groups. ** denotes significance at 5% level. Figures in () represent negative returns.*

The results of the non-financial firms strongly support the results from the full sample and indicate that including financial firms weakened the signal. For the Strong firms, the average return increases from 25.18% to 29.29% when we exclude financial firms, and it is still statistically significant at 5% ($t = 2.28$, $p = 0.044$). Even better is that we maintain the ordering of the average returns: Strong > Neutral > Weak: 29.29% > 19.74% > 14.52%.

The Strong-Weak hedge spread increases from 10.60% to 14.77% in the non-financial sample. Although the spread fails to reach the conventional 5% significance level ($t = 1.73$, $p = 0.111$), it approaches the 10% threshold, and the sign is consistent across all but three of the 12 years. The year 2022 stands out as an exception, with the hedge spread falling to -35.30% - the Weak group contained only one stock (which had an unusual year driven by idiosyncratic factors). And that single-stock concentration severely distorted the annual spread.

An interesting fact to note is that, unlike the full sample Weak group, the non-financial Weak group has data available in all 12 years. This is because removing financial firms – many of which were in the Neutral portfolio due to the distorted effect of their business model on signals like leverage and liquidity – space opened in the lower end of the distribution for non-financial Weak firms' scores. This more reliable estimate of the baseline spread is part of why the spread t-statistic is slightly better, although still not 5%-significant.

5.2.2. Model 2

Table 9 provides the distribution of stocks based on F_Score values across the value stocks under Model 2 for both Full sample and non-financial subsample.

Table 9: Distribution of A Score across the Value Stock Universe

A_ Score	Group	Full sample	% of Full Sample	Subsample	% of Subsample
0	Weak	1	0.32%	0	0.00%
1	Weak	3	0.95%	0	0.00%
2	Weak	1	0.32%	1	0.40%
3	Weak	3	0.95%	4	1.59%
4	Weak	6	1.90%	7	2.79%
5	Weak	18	5.70%	10	3.98%
6	Weak	25	7.91%	20	7.97%
7	Weak	24	7.59%	18	7.17%
8	Weak	31	9.81%	18	7.17%
9	Neutral	33	10.44%	20	7.97%
10	Neutral	35	11.08%	28	11.16%
11	Neutral	31	9.81%	25	9.96%
12	Neutral	19	6.01%	13	5.18%
13	Neutral	23	7.28%	20	7.97%
14	Neutral	13	4.11%	16	6.37%
15	Strong	12	3.80%	14	5.58%
16	Strong	11	3.48%	13	5.18%
17	Strong	10	3.16%	11	4.38%
18	Strong	8	2.53%	5	1.99%

19	Strong	3	0.95%	3	1.20%
20	Strong	4	1.27%	2	0.80%
21	Strong	1	0.32%	1	0.40%
22	Strong	1	0.32%	2	0.80%
23	Strong	0	0.00%	0	0.00%
24	Strong	0	0.00%	0	0.00%
25	Strong	0	0.00%	0	0.00%
26	Strong	0	0.00%	0	0.00%
27	Strong	0	0.00%	0	0.00%
Total		**316**	**100.00%**	**251**	**100.00%**

Note: A_score value ranges from 0 (worst) to 27 (best). Non-financial subsample excludes all financial sector and holding company firm years (65 observations)

5.2.2.1. Model 2: Full Sample

Table 10 presents the year-by-year portfolio returns for the graduated A Score model applied to the full stock universe (n = 316 firm-years). The A Score threshold formula used is: Strong if A Score ≥ 15, Neutral if 9 ≤ A Score ≤ 14, Weak if A Score < 9

Table 10: Model 2_Full Sample: Annual Portfolio Returns by A_Score Group (2014-2025)

Year	Strong	Neutral	Weak	NIFTY 100	Value	S-W	S-100
2014	63.62	56.62	55.63	31.98	56.50	7.99	31.64
2015	20.31	8.01	(12.20)	11.49	0.10	32.51	8.82
2016	27.48	3.11	(6.44)	(0.29)	1.15	33.92	27.78
2017	45.19	31.27	33.88	16.86	34.61	11.31	28.33
2018	(12.24)	3.18	16.09	11.77	3.01	(28.33)	(24.01)
2019	2.03	11.28	15.83	8.08	11.08	(13.79)	(6.05)
2020	(19.83)	(25.28)	(26.19)	(11.74)	(24.30)	6.36	(8.08)
2021	N/A	61.19	96.22	52.18	76.01	N/A	N/A
2022	(15.49)	(5.69)	1.09	(0.19)	(7.59)	(16.58)	(15.30)
2023	34.78	24.85	44.20	19.90	31.61	(9.42)	14.88
2024	39.00	63.97	49.05	30.93	56.23	(10.05)	8.07
2025	(7.87)	(2.70)	5.08	4.53	(0.94)	(12.95)	(12.40)
Mean	**16.09**	**19.15**	**22.69**	**14.63**	**19.79**	**0.09**	**4.88**
p-value	**0.088**	**---**	**---**	**---**	**---**	**0.989**	**0.423**

*Note: Figures (except p-value) are in %. For the year 2021, there were no stocks under the strong group. ** denotes significance at 5% level. Figures in () represent negative returns.*

Model 2's full sample results lead to a clear rejection of H2. The graduated A Score fails to produce a larger Strong-Weak spread than Model 1, and the ordering of portfolio return itself is entirely reversed. The returns follow this return pattern: Weak > Neutral > Strong, which contradicts the normal characteristic of the F_Score results. The Strong-Weak spread averages out to only 0.09% – which is statistically indistinguishable from zero (t = 0.01, p = 0.989). The strong portfolio itself does not achieve significance at the 5% level (t = 1.89, p = 0.088).

There can be multiple possible explanations for this. First, the Small Strong Group is a structural consequence of the threshold development process described in Section 4.4.2. Even after lowering the threshold from the equal interval upper boundary of 18 to 15, to increase group size, only 50 firm-year observations qualify as Strong across 12 years, compared to 88 in Model 1. In 2021, no stocks at all qualified as Strong, making the spread undefined for that year. Minimum strong group size for any year is 1 (2014), making that year's Strong return reflects a single company's performance rather than a portfolio return in any meaningful sense.

Second, and more importantly, the Weak group under Model 2 contains a substantially different composition of firms than the Model 1 Weak group. The graduated scoring system penalises firms more severely for modest performance across multiple signals, capturing many firms that would be classified as Neutral under the binary model but score below 9 on the A Score. Many of the firms in NSE 100 are financially sound, large-cap companies that experience cyclical troughs – they score low because they could not surpass the year's median on multiple signals simultaneously. In the NSE 100 context, many of these firms are financially sound, large-cap companies experiencing a cyclical trough — they score low not because they are financially distressed, but because they could not surpass the year's median on multiple signals simultaneously. As cyclical recoveries materialise, they generate strong returns, thus explaining the high returns in the Weak group.

Thirdly, the taken percentile-based grading system means that roughly 1/3rd of firms scores 0 on most signals, which is by construction, as the 33rd percentile threshold defines the lowest bucket in the A_Score. Since the firms concentrated in NSE 100 are qualitatively different from Piotroski's distressed small-cap weak firms, this approach concentrates more firms in the weak group than Model 1's binary approach.

5.2.2.2. Model 2: Non-Financial Subsample

Table 11 presents the Model 2 results for the non-financial subsample (n = 251 firm years), using percentile thresholds computed specifically for the non-financial universe in each year.

Table 11: Model 2_Subsample: Annual Portfolio Returns by A_Score Group (2014-2025)

Year	Strong	Neutral	Weak	NIFTY 100	Value	S-W	S-100
2014	107.58	67.64	56.30	31.98	67.67	51.28	75.59
2015	46.46	(3.51)	(7.01)	11.49	(2.54)	53.46	34.97
2016	35.59	0.64	(0.13)	(0.29)	5.53	35.72	35.88
2017	64.19	16.68	26.59	16.86	24.98	37.59	47.33
2018	(10.62)	2.84	12.46	11.77	3.45	(23.08)	(22.40)
2019	(6.19)	(3.93)	11.13	8.08	(0.67)	(17.31)	(14.27)
2020	(6.17)	(26.72)	(21.34)	(11.74)	(22.25)	15.17	5.57
2021	35.20	80.02	101.73	52.18	87.48	(66.54)	(16.99)
2022	(23.99)	(2.00)	10.88	(0.19)	(5.66)	(34.87)	(23.80)
2023	27.98	23.12	10.14	19.90	21.86	17.84	8.08
2024	64.46	86.48	60.67	30.93	79.46	3.79	33.53
2025	(7.87)	(3.96)	3.12	4.53	(3.72)	(10.99)	(12.40)
Mean	**27.22**	**19.77**	**22.05**	**14.63**	**21.30**	**5.17**	**12.59**
p-value	**0.036****	**0.094**	**0.049**	**-**	**-**	**0.637**	**0.204**

*Note: Figures (except p-value) are in %. Excludes all financial sector and holding companies. Percentile thresholds (P33, P67) computed from the non-financial subsample distribution in each year. — indicates Strong group was empty (2015 and 2021). ** denotes significance at the 5% level. Figures in () represent negative returns.*

The subsample Model 2 results show a partial recovery. The Strong portfolio average return rises substantially to 27.22% per year, and notably achieves statistical significance at the 5% level (t = 2.38, p = 0.036). The Neutral portfolio earns 19.77%, and the Weak portfolio earns 22.05%. However, the monotonic ordering of Model 1 is not restored: the Weak group continues to outperform the Neutral group, and the Strong-Weak spread of 5.17% is not statistically significant (p = 0.637).

The improvement in Model 2's Strong portfolio after excluding financial firms is consistent with the explanation offered above: financial firms were scoring poorly on the graduated A Score in ways that distorted both the Strong and Weak group compositions. The 2021 and 2015 years remain problematic — in both cases, no non-financial value stocks score A ≥ 15, leaving the Strong group empty and reducing the usable sample for spread analysis to 10 years.

The most notable single-year outlier is 2014, where the strong portfolio under Model 2 Non-financial subsamples' returns 107.58% — but this represents a single company's return. The entire strong group's return for that year, which is derived from only stock, is in essence the firm's idiosyncratic performance rather than a systematic signal outcome. This poses a structural limitation of Model 2's strong group definition in this universe.

5.3. Hypothesis Testing Results

Table 12 presents a consolidated key finding of all four analyses, allowing direct comparison of the two models in both the full samples and the sub-samples.

Table 12: Summary Comparison: Mean Annual Return across the four samples

Analysis	Strong	Neutral	Weak	S-W Spread	p-value (Strong)
Model 1: Full sample (n = 316 firm years)	25.18%	19.18%	14.10%	+10.60%	0.029**
Model 1: Subsample (n = 251 firm years)	29.29%	19.74%	14.52%	+14.77%	0.044**
Model 2: Full sample (n = 316 firm years)	16.09%	19.15%	22.69%	+0.09%	0.088
Model 2: Subsample (n = 251 firm years)	27.22%	19.77%	22.05%	+5.17%	0.036**
NIFTY 100 benchmark	14.63%	--	--	--	--

*Note: Strong-Weak spread is calculated as the arithmetic difference of annual returns, averaged over all the years where both groups had members. All returns are the mean annual equal-weighted portfolio returns. ** = p <0.05 (two tailed test, null; mean = 0). NIFTY 100 benchmark return: 14.63% per year on average over the same study period.*

A clear verdict for the hypotheses emerges when the four analyses are looked at in unison. H1 is supported. Model 1 consistently produces the monotonic ordering: Strong > Neutral > Weak in both samples. The strong portfolio is generating significant returns and outperforms the NIFTY 100 benchmark by 10.56% and 14.66%, respectively. 10.60% hedge spread, though economically meaningful at 10.60%, is not significant at 5% level, and this can be attributed to the structural property of the NSE 100 universe rather than as a limitation of the F_Score itself.

H2 is rejected: the A_Score does not outperform the F_Score. An inverted ordering is produced in the full sample, and although the subsample shows a partial recovery, the strong-weak spread is statistically insignificant.

CHAPTER 6: DISCUSSION

6.1. The F_Score in a Large-Cap universe.

Support for H1 – Financially strong value stocks under NSE 100 earn approximately 25% to 29% annually, against a benchmark of 14.63% return from the NIFTY 100 index – confirms that the predictive power of fundamental analysis is retained even in a large-cap universe. This is a meaningful result; the large-cap universe has high analyst coverage and, therefore, information inefficiency is presumably lower than in other market-cap segments. The study tested the proposition that the F_Score framework is effective across all market capitalisation segments in India's largely traded and liquid index. The results support the choice of NSE 100 as the universe: the F_Score discriminates meaningfully between value stocks within a large-cap universe.

Strong portfolio remains statistically significant, and this suggests that all publicly available financial statement information is not reflected instantaneously in the stock prices, even at the large-cap level. This is consistent with (Tripathy & Pani, 2017) finding that high F_Score firms in India exhibit superior future performance, and adds to the growing body of evidence that fundamental analysis can generate excess returns in Indian equity markets.

One structural difference from Piotroski's original study is worth noting, and it directly explains why the hedge spread does not achieve statistical significance despite being economically meaningful. In his US sample, the strategy's power came from both ends of the distribution: high F_Score winners and low F_Score losers both contributed to the hedge return. In the NSE 100, the Strong portfolio delivers consistent alpha. The Weak group, however, is characteristically thin in this universe — a finding that reflects the quality composition of large-cap indices rather than any deficiency in the signal. Firms in the NSE 100 with low F_Scores are predominantly cyclical heavyweights in steel, refining, and commodities experiencing a sector trough, and they often recover strongly in subsequent periods. The hedge spread, therefore, has limited statistical power in this universe because the index's composition reduces the population of severely financially deteriorated firms, which is precisely the type of firm the short leg of a Piotroski strategy requires.

6.2. Why the A_Score did not outperform the F_Score

The failure of Model 2 to outperform Model 1 in the full sample raises important questions about the conditions under which graduated scoring might be expected to add value. Several mechanisms contributed to Model 2's underperformance.

A given absolute level of financial performance can translate into different scores in different years, with the year-specific percentile graduation. A firm with an RoA of 6% can get a score of 3 in a particular year when the universe median RoA is 4%, but also can score only 1 in a particular year when the median shoots up to 9%. This introduces cross-year incomparability: the score reflects relative performance within the year's cross-section rather than an absolute standard of financial health. Contrastingly, Model 1 has a binary threshold emphasising only the direction: an absolute standard with stable meaning across years. In a universe where overall metrics improve over time, this stability is particularly important: absolute standards maintain their meaning while fluctuations can be seen in the case of relative ranking.

Model 2's strong group is excessively concentrated, touching as few as one stock in some years. This means that this group's annual return in those years reflects a single firm's idiosyncratic returns rather than a systematic effect, thereby introducing extreme volatility, potentially obscuring the underlying signal. The broader strong threshold ($F \geq 7$) typically contains 8 to 14 stocks per year, which provides sufficient diversification for portfolio-level returns to be meaningful.

Model 2 recovering partially after excluding financial firms suggests that the A_Score is sensitive to the presence of financial companies, whose balance sheet elements are structurally different from those of other sectors. This causes them to score unusually on leverage, liquidity and asset turnover in ways that were not tuned to their business model. Extreme values in the percentile distribution shift the entire scoring calibration, affecting all firms' scores in a particular year. A homogeneous distribution is created after removing these firms, wherein the graduation scoring works cleanly.

6.3. Comparison with Prior Indian Market Evidence

Model 1 results are generally consistent with (Tripathy & Pani, 2017), wherein the high F_Score firms in the NSE top 500 outperform future performance measures. This study's findings extend their work to the NSE 100 universe, and the results hold good even in a concentrated large-cap universe.

The findings of this study indicate higher levels of statistical significance for raw returns as compared to the findings of (Tripathy & Pani, 2017) who noted that the relationship was prominent for future valuation multiples. The difference in findings may be attributed to the current study's longer period, the adaptation of a different methodology, with a specific focus on value stocks rather than the entire broad NSE 500 companies.

(Sharma et al., 2021) findings suggested that modified scoring can improve returns. However, the modifications proposed were for changing the relative weights of specific signals based on their economic importance. The current study looks at a different kind of modification, wherein the reweighting is based on the improvement size. These are conceptually distinct modifications.

CHAPTER 7: CONCLUSIONS & RECOMMENDATIONS

7.1. Key Conclusions

The purpose of this study was to determine if a reweighted version of Piotroski's F_Score could increase the mean annual return earned by a value investor operating within the NSE 100. Using a sample period of 12 years, covering from July 2013 to June 2025 and 189 unique NSE 100 constituents, two models were tested: the original F_Score (Model 1) and the graduated scoring framework, A_Score (Model 2), wherein each signal is awarded 0 to 3 points based on year-specific percentile performance.

H1 is supported, with the F_Score successfully identifying financially strong value stocks within the NSE 100 that generate returns significantly above benchmark returns. The strong portfolio earns a mean annual return of 25.18% and 29.29% in the full sample and the subsample, respectively, which are both significant at 5% level. The predicted monotonic ordering of: Strong > Neutral > Weak is confirmed in both samples. Hedge spread averages out to 10.60%, though economically meaningful, it is not statistically significant, a result that can be attributed to the structural composition of NSE 100. Long-only strong portfolio is performing better; it is unambiguous and represents a practically actionable insight for NIFTY 100 benchmark investors.

H2 is rejected. The graduated F_Score (A_Score) fails to surpass the binary F_Score and results in a reversed ranking within the entire sample. This results in an excessively small strong group, introduces cross-year comparability issues and is susceptible to distortion from the presence of financial sector firms. Although the non-financial Model 2 shows some recovery (Strong portfolio $p = 0.036$), the monotonic order is not reestablished, and the hedge spread remains statistically insignificant. These results indicate that for the NSE 100 universe, a straightforward, absolute binary threshold provides more reliable and interpretable outcomes than a complex graduated scoring system.

7.2. Managerial Insights

For practitioners, the clear recommendation is to focus on NSE 100 value stocks with F_Score ≥ 7. This simple filter has delivered superior returns above the NIFTY 100 of about 10% to 15% per year on average. The graduated scoring does not appear to justify its added complexity in improving return outcomes in this universe.

7.3. Implications for the Industry

The study highlights a unique aspect of NSE 100, which sets it apart from Piotroski's initial US small-cap scenario: the index's focus on large-cap stocks results in a scarcity of companies facing severe financial distress. This scarcity, however, reduces the statistical strength of the hedge portfolio, but it does not lessen the signal's ability to successfully separate robust firms. In the context of NSE 100, this F_Score framework is primarily based on a long-only strong portfolio approach rather than a long-short hedge.

7.4. Limitations

The present has its set of limitations that should be acknowledged in interpreting the results. Firstly, the sample period covers only 12 years, which covers multiple market cycles: the 2015-2016 market correction, the 2020 COVID-19 crash, 2020 post-COVID recovery. The study period is shorter than Piotroski's 21-year study period. This limits the statistical power for the time series t-tests, particularly for the hedge spread analysis.

Secondly, there is an absence of formal risk adjustment, which means that the superior returns reported may reflect compensation for unmeasured risk exposures that are not included in the NIFTY 100 benchmark. Value premium, a known risk factor, along with the F_Score, could be reflected in the high BM strong portfolio. Separating these two would require a regression framework such as Fama-French's three-factor models, which was not adopted here.

Thirdly, the analysis assumes that there are zero transaction costs, zero impact costs and immediate execution at June 30 prices. However, in practice, large-cap stocks have low impact costs, but this assumption overstates the achievable net return, especially for smaller stocks within the weak group.

Fourthly, the inclusion of financial sector companies within the full samples creates interpretative challenges for many F_Score signals, as discussed throughout the paper. The robustness check for the non-financial subsample addresses these issues, but there is still scope for judgment calls that can be taken on what constitutes NBFCs, insurance and diverse financial companies.

7.8. Scope for Future Research

Future research could extend this study by examining the F_Score's performance in the NSE midcap universe, where the distressed-firm tail is more populated, and the hedge strategy may be more statistically significant. Additionally, a sector-specific modified scoring system can allow meaningful application of the framework across the full NSE 100, which could specifically address the presence of financial sector firms in the universe, without excluding the financial sector.

REFERENCES

Fama, E. F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, *25*(2), 383. https://doi.org/10.2307/2325486

Fama, E. F., & French, K. R. (1992). The Cross-Section of Expected Stock Returns. *The Journal of Finance*, *47*(2), 427–465. https://doi.org/10.1111/j.1540-6261.1992.tb04398.x

Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, *33*(1), 3–56. https://doi.org/10.1016/0304-405X(93)90023-5

Fama, E. F., & French, K. R. (1998). Value versus Growth: The International Evidence. *The Journal of Finance*, *53*(6), 1975–1999. https://doi.org/10.1111/0022-1082.00080

Hyde, C. E. (2013). An Emerging Markets Analysis of the Piotroski F Score. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.2274516

Piotroski, J. D. (2000). Value Investing: The Use of Historical Financial Statement Information to Separate Winners from Losers. *Journal of Accounting Research*, *38*, 1. https://doi.org/10.2307/2672906

Porta, R. La, Lakonishok, J., Shleifer, A., & Vishny, R. (1997). Good News for Value Stocks: Further Evidence on Market Efficiency. *The Journal of Finance*, *52*(2), 859–874. https://doi.org/10.1111/j.1540-6261.1997.tb04825.x

Sharma, D., Kapur, S., Bhatnagar, S., & Singh, T. (2021). Modified piotroski score for higher returns. *Asian Journal of Research in Banking and Finance*, *11*(1), 1–7. https://doi.org/10.5958/2249-7323.2021.00001.8

Tikkanen, J., & Äijö, J. (2018). Does the F-score improve the performance of different value investment strategies in Europe? *Journal of Asset Management*, *19*(7), 495–506. https://doi.org/10.1057/s41260-018-0098-3

Tripathy, T., & Pani, B. (2017). Effect of F Score on Stock Performance: Evidence from Indian Equity Market. *International Journal of Economics and Finance*, *9*(2), 89. https://doi.org/10.5539/ijef.v9n2p89

Walkshäusl, C. (2020). Piotroski's FSCORE: international evidence. *Journal of Asset Management*, *21*(2), 106–118. https://doi.org/10.1057/s41260-020-00157-2